

Recurrent Memory: Learning from Demonstrations Without Growing the State

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Central claim. Adding demonstrations can be absorbed into a fixed-size recurrent state, but overlapping evidence can interfere and cause retrieval errors.

In-context learning asks a simple architectural question: what carries demonstrations forward as computation proceeds? A recurrent-memory model compresses the stream into an evolving state that is repeatedly read and updated. Unlike a token-by-token history, the state shape can remain fixed as the sequence grows, but compression can create interference and forgetting.

Falsifiable claim: adding demonstrations can be absorbed into a fixed-size recurrent state, but overlapping evidence can interfere and cause retrieval errors. The artifact uses a deliberately small 5×5 associative-memory substrate. A demonstration provides a key vector k and value vector v ; the toy writes an outer product into a fixed matrix with decay: $M_t = \lambda M_{(t-1)} + \alpha k_t v_t^T$. A query reads $q^T M_t$ and returns scores over candidate values. The learner can add examples without increasing the matrix dimensions, then increase key similarity or add a conflicting example to make crosstalk visible beside ground truth.

The computation is a **synthetic teaching toy**, not an official BDH or BDH-CQ implementation. Its role is to expose state, write, read, capacity and interference as visible variables. The narrow experiment is deliberately stronger than a static definition because a learner can change one variable and test whether the predicted consequence appears.

The BDH connection is substantive but carefully bounded. The Dragon Hatchling paper describes a scale-free graph of locally interacting neuron particles and states that BDH working memory during inference relies on synaptic plasticity with Hebbian learning. It also reports sparse, positive activations and emphasizes interpretability of state. This motivates the idea of memory as evolving internal state, while the official architecture is much richer than the 5×5 toy.

BDH-CQ makes the selected in-context-learning topic especially direct. Its August 2026 technical report describes inference-time inputs continuously updating recurrent memory; the query is then solved through iterative computation in a high-dimensional latent space without verbalizing intermediate reasoning. The correct teaching bridge is therefore: **demonstrations update state -> state supports the query -> latent computation can refine the answer**. The toy teaches the first two pieces and uses the paper for the third, without claiming behavioral equivalence.

The idea is active across current sequence-model research. Griffin (2024) combines gated linear recurrences with local attention; RecurrentGemma (2024) applies the Griffin design in open language models and highlights fixed-sized state; Titans (2025) introduces learned neural long-term memory for historical context. These works, together with BDH and BDH-CQ, show recurrent state and learned memory as active post-Transformer design directions, not merely historical RNN ideas.

The main limitation is unavoidable: fixed state is not infinite memory. Similar inputs can compete for the same representational directions, while decay can both preserve capacity and erase useful evidence. The toy does not establish benchmark-level reasoning, and recurrent memory is not automatically 'reasoning'. A useful learner outcome is more precise: explain what stayed fixed, what changed, why the estimate moved, and how that mechanism resembles - but is not identical to - recurrent memory in BDH-CQ.

Representative research comparison

System / work	Core memory idea	Why it matters here
Griffin / RecurrentGemma (2024)	Gated recurrent state + local attention	Concrete modern recurrent sequence modeling with fixed-size state.
Titans (2025)	Learned neural long-term memory	Shows active work on storing long past context in learned memory.
BDH / BDH-CQ (2025-2026)	Synaptic or recurrent latent memory	Direct Pathway link: evolving inference-time state stores and supports new information.

Evidence discipline: the interactive substrate is synthetic and educational. It demonstrates the mechanics of a recurrent associative state, not an official BDH/BDH-CQ checkpoint. Claims about BDH and BDH-CQ are sourced to the primary papers above.

[1] Kosowski, A. et al. (2025). The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. arXiv:2509.26507. <https://arxiv.org/abs/2509.26507>

[2] Engdahl, B. et al. (2026). BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. arXiv:2608.09888. <https://arxiv.org/abs/2608.09888>

[3] De, S. et al. (2024). Griffin: Mixing Gated Linear Recurrences with Local Attention for Efficient Language Models. arXiv:2402.19427. <https://arxiv.org/abs/2402.19427>

[4] Botev, A. et al. (2024). RecurrentGemma: Moving Past Transformers for Efficient Open Language Models. arXiv:2404.07839. <https://arxiv.org/abs/2404.07839>

[5] Behrouz, A., Zhong, P., Mirrokni, V. (2025). Titans: Learning to Memorize at Test Time. arXiv:2501.00663. <https://arxiv.org/abs/2501.00663>

[6] Pathway (2026). The Equations of Reasoning. <https://pathway.com/research/the-equations-of-reasoning>